VIDIKA MIRPURI
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EDUCATION

The Cathedral and John Connon School, Mumbai, India; 2014 – Present
– International Baccalaureate Diploma Program, Grade 12; 2025 – Present
– ICSE Board Examinations, Grade 10 2022 – 2025

STANDARDIZED TESTS

- **SAT:** Total – 1540; Reading and Writing– 750; Math – 790 2026
- **IELTS:** Overall – 8.5 2026

ACADEMIC HONORS AND AWARDS

- Distinction (Honour Roll) every year from Grade 8 to Grade 12 2022 – 2026
- Gold Award in the Queen's Commonwealth Essay Competition 2023
- Crest Gold Award from the British Science Association for "Visionless Venture" 2025
- Gold Honour in the International Physics Competition, top 1% of participants 2025
- Silver Award in the Global Quantum Mechanics Challenge, top 1% of participants 2026
- Distinction in the International Chemistry Quiz 2026
- Very High Commendation in the John Locke Global Essay Competition (Theology) 2026
- 1st place in Integration round and 2nd place in Algebra round of the Integrated Math Tournament 2026
- Distinction in the Sir Isaac Newton Physics Competition – University of Waterloo 2026
- Asian Youth Engineering Physics Olympiad – Top Talent Award 2026

PHYSICS RESEARCH

Research under the mentorship of Dr. Karishma Desai February 2026 – Present

Structural Equation Modelling of Detector Response in Hadron-Initiated Extensive Air Showers

- Reviewed the literature on extensive air showers and detector response; designed a 15-item Likert-scale questionnaire and collected 98 responses.
- Modelled three latent constructs (soil-absorbed muon detection, RF antennas, hybrid measurement) with PLS-SEM testing reliability, AVE, HTMT, Fornell-Larcker, SRMR and statistical power.
- Published in the Journal of Emerging Technologies and Innovative Research http://www.jetir.org/papers/JETIR2610157.pdf

Independent Researcher June 2026 – Present

Faraday Rotation: CIRADA Rotation Measure Catalogues

- Cross-matched the CIRADA RMTable to get 4,222 pairs of the same radio source, and fitted a mixture model against observing frequency
- Found that catalogues disagree far more than their quoted uncertainties imply, with the gap widening at higher frequency.
- Published in the International Journal of Physics Research and Development https://iaeme.com/MasterAdmin/Journal_uploads/IJPRD/VOLUME_7_ISSUE_2/IJPRD_07_02_002.pdf

Independent Researcher April 2026 – Present

Local Mass Density from Hipparcos K Giants

- Built a Bayesian Jeans-Poisson model of K giants from the public Hipparcos catalogue, using MCMC and simulated data to validate it
- Measured the local mass density and found that that Hipparcos data alone is too uncertain to settle the dark-matter question.
  https://ijnrd.org/papers/IJNRD2610172.pdf

Research Intern at the STAR Lab, Indian Institute of Technology, Bombay June 2026

Worked under Dr. Varun Bhalerao, Ph.D. in Astronomy and Astrophysics (California Institute of Technology)

- Investigated Gamma-Ray Burst physics, including the Compactness Problem, relativistic outflow models and high-energy emission mechanisms, and presented the findings in a technical research presentation
- Performed machine-learning screening for GROWTH-India Telescope observations, classifying 200+ potential sources and separating them from background noise and instrumental artifacts

Online Courses:

- Understanding Einstein: The Special theory of Relativity, final grade of 97.51%, Stanford
- The science of the Solar System, final grade of 90.38%, Caltech

LEADERSHIP INITIATIVES

February 2026 – Present

Founder – Little Miss Momentum, Indian Women Scientists Association (IWSA)

A physics outreach initiative with IWSA that brings mechanics and astrophysics to young girls and teachers through low-cost, hands-on learning.

- Designed a series of hands-on physics toys (DC Motor, Hovercraft, etc) from scrap and recycled materials, translating motion, force, energy and momentum into experiments that can be built from daily-use materials
- Planned astrophysics demonstrations teaching how the GROWTH-India Telescope detects transients, linking classroom physics to real research
- Trained 50+ teachers from Mumbai's government middle and secondary schools to build and demonstrate the toys, so the programme reaches students beyond my own sessions
- Authored a pamphlet documenting the toys, making the programme easy to replicate

Founder, Developer and Designer - Astrotots August 2026

An AI-powered educational platform that turns any physics or astrophysics concept into a playable game for children aged 6–16 worldwide.

- Built the platform end to end: coded, designed and deployed the full website
- Integrated AI to generate a custom interactive game in under a minute from any physics concept a child types in, alongside built-in modules
- Designed each game with mission objectives and explanations of the physics and astrophysics concept, adjusted to the child's age range

https://astrotots2-43xatfctj-n-07fc.vercel.app/

Head of STEMminists, The Cathedral and John Connon School 2025 – Present

A student-led outreach club bridging the gender gap in STEM by teaching middle school girls through hands-on science.

- Lead a school club of 50 members focused on closing the gender gap in STEM among middle school girls
- Design and run weekly science sessions, turning complex STEM topics into engaging experiments and guided worksheets

Founder, Developer and Designer – Visionless Venture 2024 - Present

A pair of accessible, voice-first apps for blind and visually impaired users, built on Gemini AI. Awarded the CREST Gold Award by the British Science Association.

- Built a transit app for the blind that gives real-time platform numbers and arrival times from spoken travel details, with voice-recorded requests, audio responses and directional guidance from signs to the correct platform inside the station
- Built a fashion app that analyses outfit selfies and gives detailed audio feedback on style coordination, and suggests personalised outfits from photos of the user's closet, with the exact location of each item
- Designed custom Gemini-based workflows for image recognition, voice input and context-aware feedback across both apps

Founder and Developer – Ahead of the Curve August 2026

A forecasting tool based on my Galactic Dynamics research paper, predicting how survey design affects Galactic mass-density estimates.

- Built a forecasting model using statistical modelling and coding, calibrated on 68 MCMC simulations
- Predicted how sample size, radial-velocity coverage, parallax errors and survey completeness affect measurement precision
- Helps researchers choose sample sizes and radial-velocity coverage
  https://vidikaunicorn-max.github.io/Ahead-of-the-Curve/

STEM Experiences

Intern, Medtronics

- Studied the capture management algorithms of the world's smallest pacemaler, Micra (93% smaller than traditional pacemakers), which track the capture threshold and adjust pacing output so each pulse reliably depolarises the myocardium without wasting battery
- Analysed the physics of miniaturization: how shrinking the device and electrode changes electric field distribution, current density and energy delivered per pulse 2025

Intern, Teksons

- Analysed radiator and combi-cooler manufacturing through thermodynamics and heat transfer, covering conduction, convection and heat dissipation
- Linked industrial design choices (such as fin geometry, material selection and the flow of coolant and air) to the physical principles behind them 2026

Observership, Greaves Cotton 2026

- Studied the physics of electric mobility: how LFP batteries store and deliver energy, how motors convert it into torque, and how IoT connectivity monitors performance
- Compared powertrain technologies (electric, CNG, hydrogen and Euro V+ engines) in terms of energy conversion, efficiency and emissions

Research Intern, Vanderbilt University

Mentor: Dr. Carcia Carson

- Designed a nanoparticle-based osteosarcoma vaccine from first principles, applying biophysics and nanoscale transport physics to model lipid nanoparticle delivery dynamics and tumor targeting of osteosarcoma stem-cell antigens
- Combined computational modelling and systems engineering to optimise antigen delivery and immune activation pathways

ART PORTFOLIO

Artist, Mythematics - An Art Series Fusing Mathematics and Indian Mythology 2024 – Present

- Created a series of 8 paintings fusing abstract mathematics with Indian mythology, each built around a different mythological creature
- Expressed the paradox of two worlds: the exact logic of mathematics against the symbolism of myth, pairing the Riemann surface and zeta function with figures like Matsya, the fish avatar, and Gandaberunda, the two-headed bird

MODEL UNITED NATIONS 2022 – Present

Secretariat Member, Cathedral Model United Nations

- Chaired the British House of Commons committee at CMUN 2025, the largest committee at the largest MUN in Asia
- Participated in 10+ MUN conferences and won 8 awards

VOCAL TRAINING 2020 – Present

- Trained in Hindi classical and contemporary rock and pop vocals
- Completed the Trinity Rock and Pop Vocals exams through Level 7, scoring Distinction
- Performed at several shows to raise money for charity